

Software Requirements Specification

KAMA UNIVERSITY

v3.0.0

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Revision History

Name	Date	Reason For Changes	Version
Ava McIntosh	10/28/2025	Added prerequisites to courses in the database.	2.0.0
Morgan Hunt	11/2/2025	Initial work on schedule generation.	2.0.0
Kalli Koppin	11/3/2025	Merged database folder into main folder and added the student database connection.	2.0.0
Ava McIntosh	11/4/2025	Combined all sql files into one databaseScript.	2.0.0
Kalli Koppin	11/4/2025	Fixed student display for multiple programmed students.	2.0.0
Ava McIntosh	11/5/2025	Added functions to change student variables and to add / remove users.	
Morgan Hunt	11/6/2025	Added to the schedule generation to take in input for semesters.	
Kalli Koppin	11/9/2025	Updated functional and technical requirements in index.html.	
Morgan Hunt	11/12/2025	Set up algorithm for schedule generation to be tested.	
Ava McIntosh	11/13/2025	Removed Non-Psy Electives, changed base credit hours for database.	
Morgan Hunt	11/13/2025	Finalized schedule generation.	
Kalli Koppin	11/18/2025	Added song to loading page.	
Kalli Koppin	11/18/2025	Added a course page where the user can search through all offered courses.	
Morgan Hunt	11/20/2025	Schedule generated for student gets stored into the database.	
Kalli Koppin	11/25/2025	Updated app and added ability to pull faulty data from database.	
Ava McIntosh	11/25/2025	Added functions to remove and add a course.	
Kalli Koppin	11/29/2025	Added preference buttons for student schedule.	
Morgan Hunt	12/1/2025	Schedule Co-Op generation patch.	
Kalli Koppin	12/2/2025	Added fail-cases for schedules not able to be generated.	
Morgan Hunt	12/3/2025	Patch to handle null values in schedule generation.	

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1. Introduction

1.1 Purpose

This software was built to provide universities with a simple and efficient way to track and manage schedule generation for their students.

1.2 Document Conventions

Convention	Description
Formatting	Heading 1 - Used for major section headings. Heading 2 - Used for subsections inside of major sections. Link - Used to connect other website / documents
Terminology	“User” - refers to students and staff that can interact with the application “System” - refers the software application
Requirements	All requirements are stated in the Requirements document. Requirements are defined by REQ-1:

1.3 References

Project Repository: <https://github.com/WSU-kduncan/KAMA-University>

Static Page (GitHub Pages): <https://wsu-kduncan.github.io/KAMA-University/>

Requirements Spreadsheet: [Requirements3.xlsx](#)

2. Overall Description

2.1 Product Perspective

This product is an add-on application for existing university systems. It provides students and staff with a simple platform to create, manage, and view schedules. The primary purpose of this system is to simplify schedule generation for students. A student will input their preferences, and a schedule will be generated based on the values chosen. This product will not replace existing university applications but will be an addition to previous systems. This application will make scheduling easier and more intuitive for users.

2.2 Product Features

Major Feature	Description
Schedule Generation	Generates a schedule for a student based on their preferences.
Course List	Displays all courses offered. Users can search for specific courses based on name, credit hours, department, etc.
Log In Page	Users must log in with a password and username to access the application.
User Display	Displays important information about the current user.
Preference Modification	Allows a student to modify their schedule preferences.
User Privileges	Users have access to different data and functions based on their privileges. EX: Admin can remove a course from the course list but a Student and Advisor can only view the course list.

2.3 User Classes and Characteristics

2.3.1 Students

Students use the system to review academic progress, update their schedule preferences, add or remove co-ops, add or remove summer courses, and set the amount of credit hours per semester. Characteristics for students include low to moderate technical expertise, occasional use of the system, clear navigation, helpful error messages, simplified workflows, and restricted access to only their personal academic information.

2.3.2 Faculty Advisors

Faculty members who advise students and use the system to generate schedules, review program requirements, and advise student academic plans. Characteristics include moderate to high technical proficiency, frequent use of the system, access to multiple student information boards, and must be able to rely on accurate program logic.

2.3.3 System Administrators

Students use the system to review academic progress, update their schedule preferences, add or remove co-ops, add or remove summer courses, and set the amount of credit hours per semester. Characteristics for students include low to moderate technical expertise,

occasional use of the system, clear navigation, helpful error messages, simplified workflows, and restricted access to only their personal academic information.

2.4 Operating Environment

2.4.1 Hardware Environment

The system shall run on any standard computer device such as a desktop and laptop. No specialized hardware is required beyond a device capable of running a web browser.

2.4.2 Software Environment

The system shall utilize Python 3.0 with Flask framework and JavaScript for backend functionality. The frontend of the system shall utilize HTML5 and CSS3. The database shall use MariaDB/MySQL and DBeaver. For version control and a static team website, GitHub and GitHub pages will be used. For deployment of the Degree Admin system, the system shall be used on a local Flask development server.

2.4.3 Supported Operating Systems

The system must operate correctly on major operating system platforms such as Windows 10 or later, macOS, and Linux.

2.4.4 Browser Requirements

The system requires a modern, standards-compliant browser with JavaScript enabled, cookies enabled, HTML5 and CSS3 support. Compatible browsers include Google Chrome, Firefox, Microsoft Edge, and Safari.

2.4.5 Coexisting and Interacting Software

The system must coexist with a Flask runtime environment and a database server.

2.5 Design and Implementation Constraints

Currently there is no way to change a student's major, minor, username, or password. This will limit the number of changes that can be made without having to go straight into the database to change it. Another limitation is the database is hosted locally from a SQL script, meaning changes made on each system will not transfer to another system.

2.6 Assumptions and Dependencies

We are assuming that MariaDB and DBeaver will continue to be free and useable. If either of them stops being updated or can no longer work, the entire system will no longer work. The scalability is another unknown. We have the capacity to add new classes and delete classes but currently there is no way to remove and add users so it would be possible but not sure how many would be able to be added.

3. System Features

3.1 Degree Admin Authentication Page

3.1.1 Description and Priority

When a user first opens the degree admin website, they will be presented with a login page. Here they can enter a username and password for one of the saved users in the system. This is a HIGH priority item.

3.1.2 Stimulus/Response Sequences

User enters authentication page. The system displays a box for the username, a box for the password, and a login button to enter this information.

User enters username in username box. The system displays this username as it is being typed in the box.

User enters password data into the password box. The system does not display the password to the user. It instead displays a number of placeholder characters for however long the password is.

User presses the "Login" button which sends the username and password data through the system to the authentication algorithm which checks the data entered against the saved data in the system. If the given data does not match any of the data in the system, the user cannot get past this page, otherwise the user is directed to whatever page is related to the user that they entered.

3.1.3 Functional Requirements

AUTH-1: The User will see the authentication when opening the degree admin website for the first time. This site has a username field, a password field, and a login button

AUTH-2: The user will enter a username into the username field. This field is not checked until the login button is pressed.

AUTH-3: The user will enter a password into the password field. This field is not checked until the login button is pressed.

AUTH-4: The user will press the login button. If the user entered data that does not match another user in the system, then they will be told the login information is invalid and prompted to enter information again. If the data is valid, then they will be taken to a new page related to the user type of the data they entered. These user types are Student, Faculty, and Admin.

3.2 Degree Admin Student User Page

3.2.1 Description and Priority

After a user logs into the website, they will be taken to a page that correlates with the user type of the user data that was entered into the authentication. If the user is a student they will be taken to this page. Student's can navigate this page to generate a

schedule, modify preferences for that schedule, view their information, and view all available courses. This is a HIGH priority item.

3.2.2.1 Stimulus/Response Sequences

User enters page. In the middle of the page there is a large box for a schedule. If there is already a schedule generated in the system, the schedule will appear here, if not it will say there is currently no schedule and that the user can generate a schedule using the generate button which is located below the schedule box. Also located below the schedule box is the preferences box. In the top left corner of the page is the student information tab. The top right corner has a button with the username.

User presses the “generate schedule” button. The system takes their stated preferences (DEFAULT 4 years, 18 credit hours per semester, no summer semesters, no coops) and generates a schedule based on their chosen majors and minors. If this schedule is impossible with the given constraints then it send an error requesting the user to adjust their preferences and regenerate. After a valid schedule is generated, it is displayed on the website.

User presses arrow buttons under generated schedule. These buttons are for navigating the schedule. The user can press the right arrow to go forward in the schedule, and the left button to go back. The arrows do not go past the range of data in the schedule.

User presses the change preferences button. This button opens a list of things they can change. The user can change the number of credit hours they are taking per semester. They can also change how many years long their program is, how many coops they are taking, and if they are taking summer semesters. These preferences are saved in the system once the save button is pressed. Years, semesters, and coops cannot be a value less than zero, and there cannot be more than 3 coops.

User presses the options button (three lines) in the top left corner. The system pulls their student information and opens a small window with that information in it. There is also a view all courses button. There are no user modifications that can be made to this page.

User presses the all courses button. The system pulls all the course data from the database and brings the user to a new website page. (Page details listed in its own section.)

User presses username button. The system logs them out and returns the user to the login page.

3.2.3 Functional Requirements

STUD-1: The user can view their schedule in the middle of the page if they have a schedule in the database. If they have no schedule in the database the middle window will say “no schedule” and suggest the user to generate a schedule.

STUD-2: The user can press the generate schedule button. This triggers an algorithm that takes all the student’s preferences and the programs they are registered in as input and generates a schedule that will meet all the requirements of their given program. If a schedule cannot be generated within the parameters and constraints, it will send an error suggesting the user to adjust their preferences and generate again.

- STUD-3: The user can press the change presses button. When pressed, the system pulls up a window where the user can input their preferences defined as number of credit hours per semester, number of years to graduate, if they want summer semesters, and how many coops they would like to do. These preferences are only saved to the database when the save button is pressed. Any data validation is done per preference during input, otherwise it is caught by the schedule generation algorithm.
- STUD-3: When the user changes the number of credits hours they take per semester. The value must be a whole number more than 0. If it is not, the value will not be saved to the database.
- STUD-4: When the user changes the length of their program in year, the number must be more than 0. If it is not, then it is not saved to the database.
- STUD-5: When the user changes the number of coops that they would like to take, the number must be a whole number between 0 and 3. The user cannot type in and save a value outside of this range.
- STUD-6: If the user has a valid schedule in the database, they can navigate through this schedule with arrow buttons located at the bottom right of the pane where the schedule is displayed. Each window of this pane displays one semester of their schedules labelled with Year (n) (Fall, Spring, Summer).
- STUD-7: The user can press the right arrow button (with a valid schedule) to navigate to the next semester in chronological order. The arrows will not display a semester that is empty or outside the range of their program. Left arrow goes back in chronological order.
- STUD-8: The user can press the three lines in the top left of the page. This will bring up a window of their student data grabbed from the database. Here there will also be a button to view the all courses page.
- STUD-9: The user can press their username at the top right of the page. This will log them out and bring them bag to the authentication page.

3.3 Degree Admin Faculty User Page

3.3.1 Description and Priority

After a user enters valid input to the authentication page, if they are categorized as a faculty user they will be taken to the faculty user page. Here faculty can view information about themselves and about their students. They can also click on the schedules of each of their students and view it in a separate page if their student has a valid schedule in the database. This is a LOW priority item.

3.3.2 Stimulus/Response Sequences

When the user enters the faculty page. The center of the page will have the information for all the students assigned to them. The system pulls the data for each of their assigned students and displays the name, programs, expected graduation, co-op, summer semesters, schedule exists (exists if there is a valid schedule in the database), view schedule button (if schedule exists). There is also a list button in the top left corner, and the username of the user is displayed in the top right corner.

User presses the view schedule button when a valid schedule exist for an assigned student. The system brings a user to a new page that shows the student's entire schedule. The user can scroll to view every semester. This button does not appear if there is not a valid schedule in the database.

User presses the top left list button. The system pulls the user information from the database and displays it in a window on the left side of the page. If there is no information for a category, it shows up as null.

User pressed their username in the top right corner. The system logs them out and takes them to the authentication page

3.3.3 Functional Requirements

FAC_1: The user can view all of the information for their students in the middle of the main faculty page. The information displayed is the name, programs, expected graduation, coops, summer semesters, if they have a schedule and a database, and a button to view that schedule if they do. If any value from the database do not exist, they will appear as NULL in the table.

FAC_2: The user can click on the view schedule button if the target student has a valid schedule in the database. IF the target student does not, this button will not appear. Upon clicking the button the user can view the entire student schedule on one page, and can navigate it by scrolling.

FAC_3: The user can press the list/options button on the top left corner of the page to view their own information from the database.

FAC_4: The user can press their username in the top right corner. This will log them out and bring them back to the authentication page.

3.4 Degree Admin System Administration User Page

3.4.1 Description and Priority

After a user enters valid login information, if their user type is admin, they will be brought to the admin page. On this page, they are able to modify and view the master course list. This is LOW priority.

3.4.2 Stimulus/Response Sequences

User enters page and can see the options/list button in the top left corner, and the user's username in the top right corner.

User presses the options in the top left corner. The user can see their own information in this window and two buttons, View All Courses and Manage Courses buttons.

User presses manage courses button. This opens a page where the user can see all the courses in the master course list and add or remove classes. To add a class, the user needs to input the Course ID, semesters offered, course code, course name, and credit hours for the course. If all this information is valid, it will be added to the database.

User presses the view all courses button and it takes them to the master course list page.

3.4.2 Functional Requirements

ADMN-1: The user can view a blank admin page with a options/list button in the top left corner and their username in the top right corner after logging in.

ADMN-2: The user can press the options/list button in the top left corner to reveal a window with their user stats. This window also has buttons to View all courses and modify courses.

ADMN-3: The user can press the Modify Courses button in the list window. This bring the user to a new page where they can see all of the courses in the database and add or remove courses

ADMN-4: The user can add a course. A new course must have a Course ID, semesters offered, course code, course name, and credit hours to be added to the database. You cannot use the same ID as another course on the list.

ADMN-5: The user can press the View all Courses button. This brings them to a page that shows the master list of courses.

3.5 Degree All Courses Page

3.5.1 Description and Priority

This page can be accessed from other user pages (admin and student), and display all of the courses in the database (The master list of courses for all programs). This is MEDIUM priority.

3.5.2 Stimulus/Response Sequence

The user can view all the courses in the database. The information displayed for each course is Course Code, Course Name, Credit Hours, Semesters Offered, and Prerequisite ID. This data is filled out by a system access to the database.

The user can search (located at the top of the list) the course list by Course Code or Course Name. As the user is entering text the system filters the course list data to only show courses that contain the text entered in either the Course code or the course name.

The user can filter data in the table at the top of the list by department of the course, semester offered, and/or credit hours. The system takes this information and only displays courses that meet all the specified filters. If a filter is not specified it treats it as allowing all courses. These filters are all pick-lists, so the user cannot enter an invalid filter.

3.5.3 Functional Requirements

CRSE-1: Upon entering the page, the user can view all the courses. The information shown for each course is Course Code, Course Name, Credit hours, Semesters Offered, and Prerequisite ID.

CRSE-2: The user can search the course by Course Code or Course Name. The Course list will display courses where the course code OR the course name contains the text entered in the search.

CRSE-3: The user can filter the list by department, semester offered, and credit hours. The courses in the list must meet the constraints of all the filters specified.

4. External Interface Requirements

4.1 User Interfaces

4.1.1 GUI Standards

All system pages follow a consistent, minimal design that uses a green accent color, chameleon iconography, and rounded UI components. The system utilizes a light theme optimized for clarity and readability across supported devices. Typography and spacing follow a uniform layout pattern across all user dashboards, ensuring a cohesive and easy-to-navigate interface.

4.1.2 Screen Layout

All authenticated pages include a persistent header bar with the KAMA logo, user greeting, notifications icon, and logout button. Some pages also include a left-side navigation menu depending on the user's role. The main content is centered on the page and displayed in simple card-style sections to maintain readability. Error and success messages appear in clearly labeled banners at the top of the screen.

4.1.3 Standard UI Components

Buttons use consistent styling such as green for primary actions like as Generate Schedule or Save Preferences, and gray for secondary actions. Form fields display labels above each input, with inline error messages appearing directly beneath incorrect or incomplete fields. Modals are used for editing student preferences and confirming certain actions, each containing clear Save and Cancel options.

4.1.4 Screens Requiring User Interfaces

The system provides interfaces for the following screens:

- Team Website
- Login Screen
- Student Dashboard
- Faculty Dashboard
- System Administration Dashboard
- Course Catalog

4.1.5 Keyboard and Accessibility Features

Users can navigate interactive elements using the Tab key, submit forms with Enter, and close modals with Escape. All icon-only buttons include ARIA labels, and the system uses semantic HTML to support assistive technologies. Color contrast levels comply with accessibility best practices.

4.2 Hardware Interfaces

The system is a web-based application and does not require direct interaction with specialized hardware. The system operates on standard computing devices capable of running a modern web browser.

4.3 Software Interfaces

4.3.1 Web Browser

The system runs in any modern browser that supports HTML5, CSS, and JavaScript such as Google Chrome, Firefox, Microsoft Edge, and Safari.

4.3.2 Backend Framework

The application uses Python (Flask) to handle routing, authentication, schedule generation, and communication with the database.

4.3.3 Database System

The system connects to a MariaDB/MySQL relational database to store student data, program requirements, course information, and generated schedules. The system exchanges structured queries and receives results in standard SQL formats.

4.3.4 Frontend Technology

The interface is built using HTML, CSS, and JavaScript, which communicate with the backend through HTTP requests and server-rendered templates.

4.3.5 Version Control & Deployment

Source code is managed using GitHub, and certain static components are deployed using GitHub Pages.

4.4 Communications Interfaces

The only requirement is a web browser. With all internal communications through the website with no other outside input other than from the database.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

5.1.1 Description and Priority

The system shall return schedule generation results within 3 seconds under normal load to ensure timely feedback for faculty and students.

5.1.2 Database Query Efficiency

Individual database queries such as schedule retrieval, programs, user information, and preference changes, shall be completed within 3 seconds upon changing.

5.1.3 Scalability

The system shall support the addition of new academic programs, course catalogs, and schedule changing rules without requiring major architectural changes.

5.1.4 Dashboard Load Time

All dashboard interfaces such as student, faculty, and system administrators shall load within 2 seconds after authentication.

5.2 Safety Requirements

5.2.1 Data Backup and Recovery

The system shall be able to be recovered in the case of corruption due to multiple back up files and be able to recover student information, faculty information, and system administration information.

5.2.2 Input Validation and Error Verification

The system shall prevent operations that corrupt student data including scheduling duplicate courses, invalid co-op placements, or negative credit hour terms.

5.2.3 Fail-Safe Generation

The system shall return a clear error message to the user if schedule generation is incomplete, corrupted, or missing preference options.

5.2.4 Restricted Modification of Academic Data

Only authorized faculty users shall be permitted to modify degree requirements, course information, or scheduling rules.

5.3 Security Requirements

5.3.1 User Authentication

All users shall authenticate using login credentials such as username and passwords. Students may only access their own records and course catalog page.

5.3.2 Audit Logging

The system shall record key events such as login attempts, schedule generation, preference edits, and academic

5.3.3 Data Protection and Integrity

The system restrict unauthorized modification of degree requirements, course properties, co-op rules, and scheduling logic to ensure academic data remains accurate and consistent.

5.3.4 Session Timeout

Use sessions shall automatically expire after 20 minutes of inactivity to protect against unauthorized access on shared or public devices.

6. Key Resource Requirements

[illegible]